

Review — Read Before You Start

Quick Reference

Percents: part = (percent as decimal) · (whole); percent change = $\frac{\text{new} - \text{old}}{\text{old}} \times 100\%$.

Lines: slope (rate of change) $m = \frac{y_2 - y_1}{x_2 - x_1}$; slope—intercept form $y = mx + b$; point—slope $y - y_1 = m(x - x_1)$.

Intercepts: x -intercept: set $y = 0$; y -intercept: set $x = 0$.

Quadratics: $ax^2 + bx + c$; vertex $x = -\frac{b}{2a}$; factoring when possible; roots solve $ax^2 + bx + c = 0$.

Pythagoras: right triangle: $a^2 + b^2 = c^2$ (hypotenuse c). Distance in the plane: same formula on $\Delta x, \Delta y$.

Mindset

Make a tiny sketch; label what is given; define your variable; check if your answer size makes sense.

Part A — Percentage Word Problems

1. A jacket's original price is \$80. It is discounted by 25% and then a 6% sales tax is applied to the discounted price. What is the final price? (Round to the nearest cent.)

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2. A population grows from 12,000 to 13,440 in one year. Find the percent increase.

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3. A laptop is on sale for \$714 after a 15% discount. What was the original price?

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4. A store marks up a \$50 item by $x\%$ to \$59. Find x .

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5. A student improved a quiz score from 72 to 90. What percent increase is this? If next quiz increases by the *same percent*, what would a 90 become?

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6. A solution is 8% salt by volume. How many milliliters of salt are in 350 mL of solution?

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Part B — Slope, Intercepts, and Graphing Lines

7. Find the slope of the line through $(2, -5)$ and $(-4, 7)$.

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8. Write the equation of the line with slope -3 passing through $(4, 2)$ in slope—intercept form.

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9. Find the x - and y -intercepts of $3x - 2y = 12$ and write the slope—intercept form.

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10. A line has y -intercept -4 and passes through $(6, 5)$. Find its slope and equation.

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11. Two lines are given: $L_1 : y = 2x + 3$ and $L_2 : y = -\frac{1}{2}x + 3$. Do they intersect, are they parallel, or perpendicular? Explain briefly.

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12. A line goes through $(a, 3)$ and $(a + 4, 11)$. Write its slope and then an equation in slope—intercept form.

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13. Sketching without a graph: Describe how you would quickly hand—graph $y = -\frac{3}{2}x + 6$ by finding intercepts and one more point. (No picture needed; list points.)

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Part C — Simple Quadratic Polynomials

14. Factor completely: $x^2 - 11x + 24$.

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15. Solve: $x^2 + 5x + 6 = 0$. Then give the x -intercepts as ordered pairs.

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16. The quadratic $y = x^2 - 6x + 5$. Find the vertex and the y -intercept.

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17. Evaluate $f(x) = 2x^2 - 3x - 5$ at $x = -2$ and $x = 4$.

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18. The product of two consecutive integers is 72. Set up and solve a quadratic to find the integers.

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Part D — Pythagoras and Distance

19. Find the length of the hypotenuse of a right triangle with legs 7 and 24.

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20. A ladder leans against a wall. The foot is 6 ft from the wall and the ladder reaches 12 ft up the wall. How long is the ladder?

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21. Find the distance between $A(-3, 5)$ and $B(9, -7)$.

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22. A rectangle is x by $(x + 3)$ with diagonal 13. Write and solve an equation for x .

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Part E — Linear Models (Concept)

23. A phone plan charges \$20 per month plus \$0.08 per text. Let T be the number of texts.
(a) Write a linear model for the monthly cost $C(T)$. (b) What is the slope and what does it represent? (c) What is the y -intercept and what does it represent?

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Label units, keep exact forms when appropriate, and check reasonableness.